Lab 4: Program Execution using GDB

Aim

To study and analyze the execution of a C program step-by-step using GDB and observe how instructions are executed in memory.

Procedure

1. Write a simple C program (e.g., addition of two numbers).
2. Open terminal and navigate to the program directory.
3. Compile the program with debugging option:

   ```
   gcc -g program.c -o program
   ```
4. Start GDB debugger:

   ```
   gdb ./program
   ```
5. Set a breakpoint at the main function:

   ```
   break main
   ```
6. Run the program:

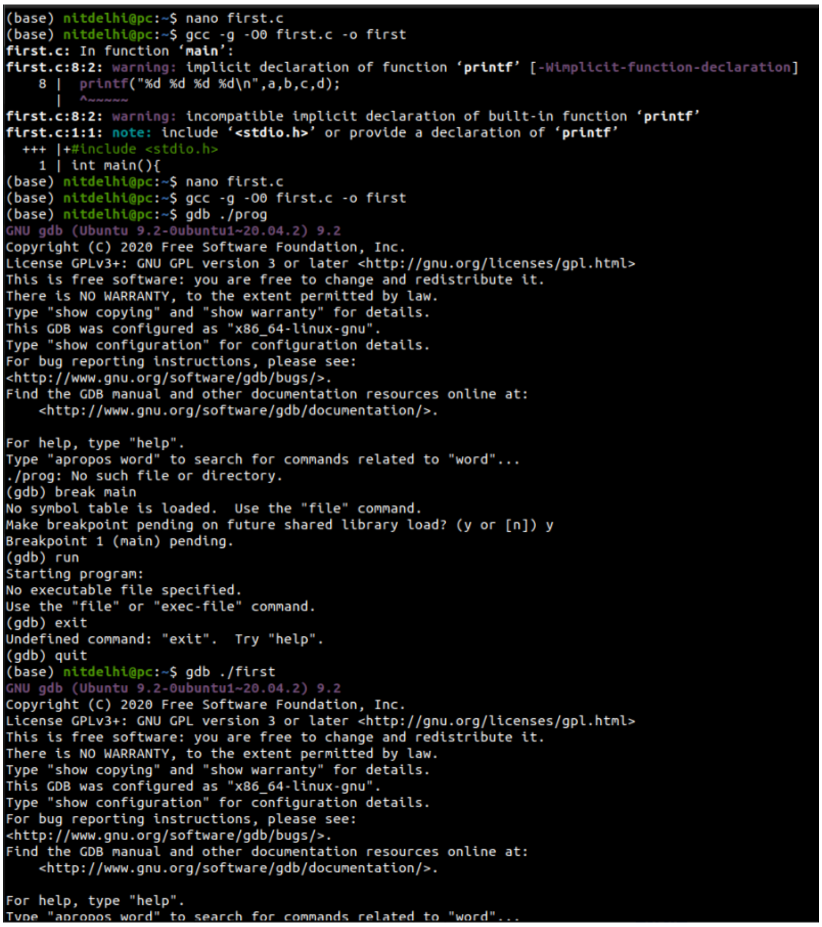
   ```
   run
   ```
7. Execute the program step-by-step using:

   ```
   next
   ```
8. Display variable values:

   ```
   print variable_name
   ```
9. View memory or registers if required:

   ```
   info registers
   ```
10. Continue execution until program ends:

    ```
    continue
    ```



```
(base) nitdelhi@pc:~$ nano first.c
(base) nitdelhi@pc:~$ gcc -g -O0 first.c -o first
first.c: In function 'main':
first.c:8:2: warning: implicit declaration of function 'printf' [-Wimplicit-function-declaration]
    8 |  printf("%d %d %d %d\n",a,b,c,d);
      |  ^~~~~~
first.c:8:2: warning: incompatible implicit declaration of built-in function 'printf'
first.c:1:1: note: include '<stdio.h>' or provide a declaration of 'printf'
  +++ |+#include <stdio.h>
    1 | int main(){
(base) nitdelhi@pc:~$ nano first.c
(base) nitdelhi@pc:~$ gcc -g -O0 first.c -o first
(base) nitdelhi@pc:~$ gdb ./prog
GNU gdb (Ubuntu 9.2-0ubuntu1~20.04.2) 9.2
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License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
./prog: No such file or directory.
(gdb) break main
No symbol table is loaded.  Use the "file" command.
Make breakpoint pending on future shared library load? (y or [n]) y
Breakpoint 1 (main) pending.
(gdb) run
Starting program:
No executable file specified.
Use the "file" or "exec-file" command.
(gdb) exit
Undefined command: "exit".  Try "help".
(gdb) quit
(base) nitdelhi@pc:~$ gdb ./first
GNU gdb (Ubuntu 9.2-0ubuntu1~20.04.2) 9.2
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License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
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For help, type "help".
Type "apropos word" to search for commands related to "word"...
```

Result

The execution of the C program was successfully analyzed using GDB.

- Step-by-step execution of instructions was observed.
- Values of variables were monitored during execution.
- The working of the program in memory was understood.

The experiment helped in understanding how programs are executed at a low level.